NEURAL ENGINEERING LAB (86-783 / 42-783 / 24-777)
INTRODUCTION LAB: CYTON SETUP

Module: Cyton Setup
Date: 8/29/2023

1) Objectives

Thoroughly read and carefully follow the instructions because this is the basics for the rest of the semester!

Familiarize yourself with your team members
Get started and familiar with the OpenBCI software
Learn to setup the software to reduce interference and noise in readings by setting custom channels
Correctly operate and get readings using the Cyton board and dongle
Work as a group to save small data files

2) Advice

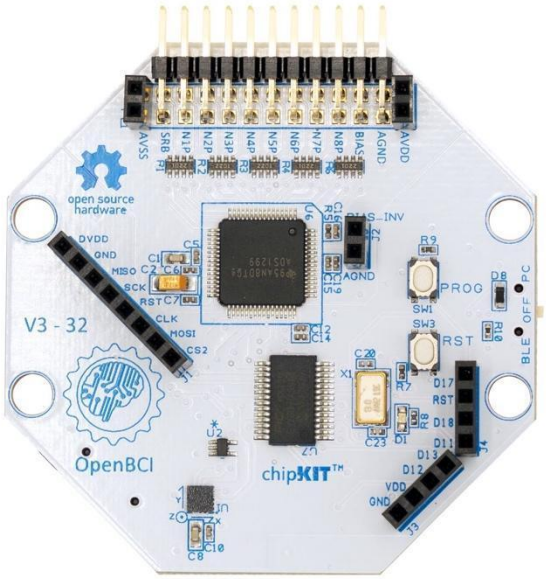
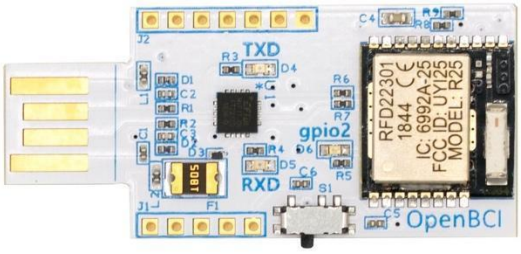
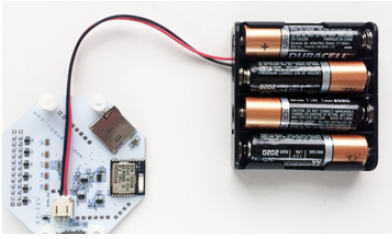
Please read through the instructions in this document thoroughly and ***complete each step before moving on***.

Key steps are highlighted in yellow, but you should read all the instructions carefully. You will want the power adapter for your laptop because the OpenBCI software and data collection can really drain your battery. In this case because the data collection is wireless it won't introduce any noise to have your laptop plugged in.

If you get stuck at any point, first debug it yourself using the Setup & Troubleshooting guide and then reach out to members of the teaching team.

Please ask for help if you are unsure about any of the steps!

3) Components and supplies

OpenBCI Cyton Board and personal laptop (with power adapter)	
Cyton Dongle (and a free USB port on your computer – you need USB A, so if you only have USB C you'll need an adapter)	
Battery pack and 4 AA batteries (or rechargeable battery pack)	
Software: OpenBCI software, MATLAB	

4) Procedures

For our recordings this semester we're going to use a hardware platform known as OpenBCI (www.openbci.com) and use their 8-channel Cyton boards to acquire various kinds of data.

⚠ Because these boards are somewhat expensive (~$1000), and because it's challenging to be both the subject and experimenter, you will work in small groups in this laboratory (2-3 people) and share a Cyton board for that group. ⚠

To get started, make sure you download the necessary software here: https://openbci.com/downloads. During this lab, you will install the software, get it to run, and see some basic signals.

A) Setting up the Cyton board

A general outline of instructions is here: (https://docs.openbci.com/GettingStarted/Boards/CytonGS/)

1) The first step is to download and install the OpenBCI GUI software **(NOTE: do not install FTDI drivers just yet; only install the OpenBCI GUI and see if it works. Read the instructions for the download of the driver and confirm if it is required for your system)**
2) Then, get a battery *(individual battery or rechargeable pack)* ready for the Cyton and plug that in *(align it with the notch to plug it in correctly)*.
3) Now, plug the USB dongle into your computer and you'll see the LED lights go on **(the switch (in black) should be set to GPIO 6 and not RESET)**.
4) Finally, turn the switch in the Cyton board to PC *(not OFF or BLE)* and a blue LED will turn on. You're now ready to connect to the Cyton board.

⚠ If you just want to get started, you can choose "Auto-Connect" from there and get started, and then click "Start Data Stream". This is the fastest way to get started, but the software will assign your Bluetooth communication channel randomly (note that the software also uses channel to mean electrode number). This doesn't work that well in a large group. So, let's go back to the beginning. ⚠

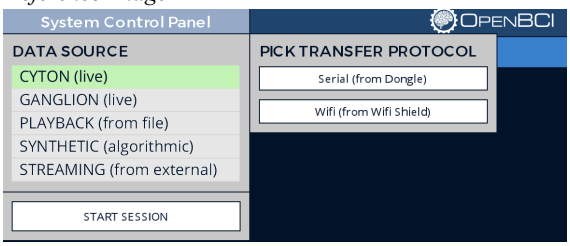
B) Communicating with the Cyton board

It's critically important that for this next step you do this one at a time.

⚠ One group should turn on their Cyton board (while all other boards are off), follow the steps below, write down their channel, and then power off their board and remove the dongle from their computer. We have to do this until all boards are configured with their own unique channels so that everyone can work together in the same room and not interfere. ⚠

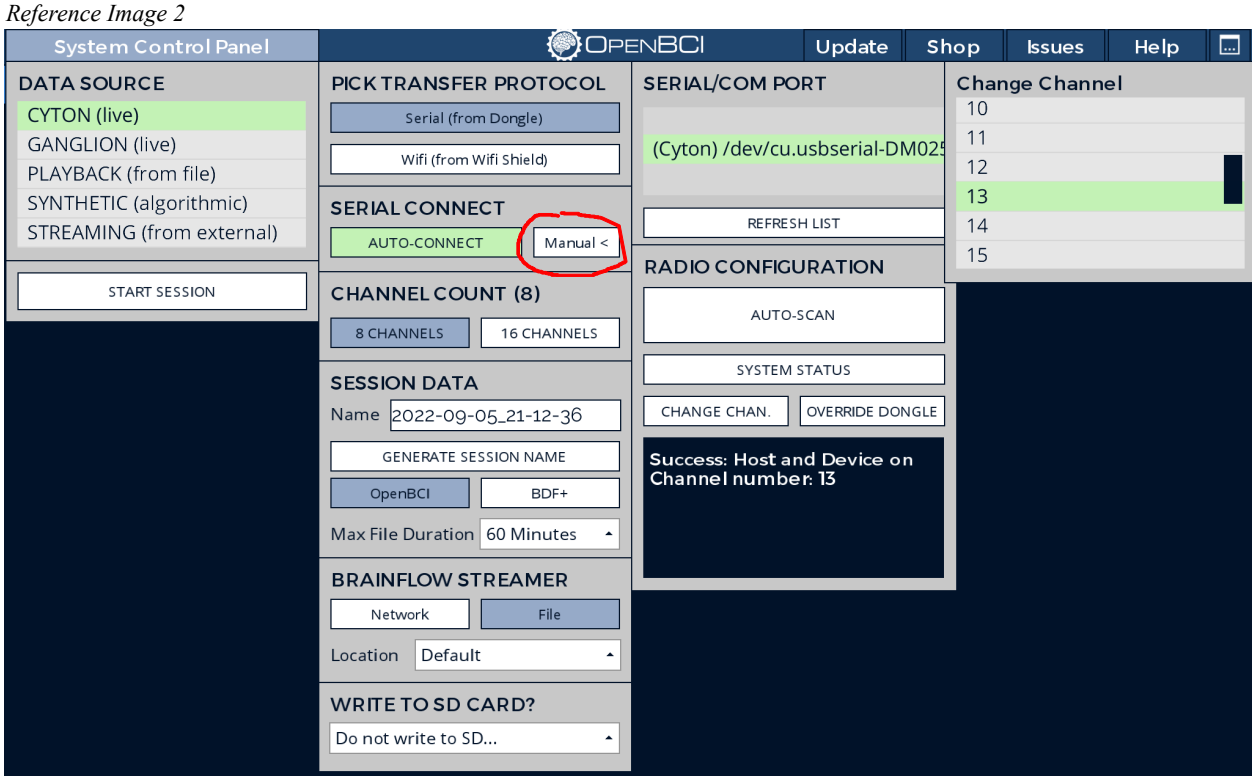
1) Open the OpenBCI software, click "Cyton (live)" → "Serial"→ "Manual *(See Reference Image 1)*

Reference Image 1



2) Select the Serial/Com port (it should be highlighted green if it's selected)
3) Click "System Status", and you should see a "success" message. If so, great! **(If not, keep trying the "Auto-Scan" button until it connects.)**
4) Check the "Channel number" mentioned in the System status message *(See Reference Image 2)*
5) Select "Change Chan." and **choose the channel the instructors have told you**.

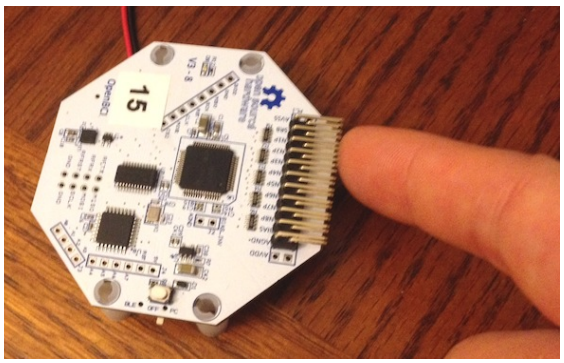
Reference Image 2

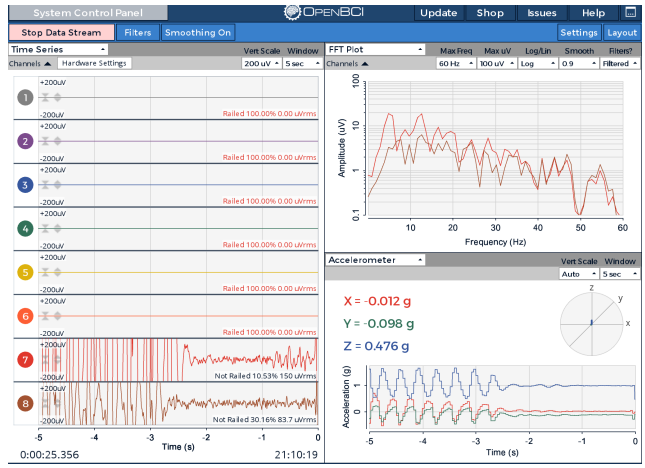


6) Click "System Status" again and ensure that the "Channel number" is the one you are assigned.
7) Disable file-saving for the "Brainflow Streamer" by changing the "Location" to "None"
8) Change the name of the session folder under "Session Data"→"Name"
9) At the top left of the app click "Start Session" and you should see the main interface.
10) Click "Start Data Stream" → tap on the Cyton gently or touch the pins on the side → Signals! **(If you see that (for example you can see the accelerometer signal if you move the Cyton board), you're ready for data collection!)** *(See Reference Image 3)*

If not, ask one of the course staff for help.

Reference Image 3





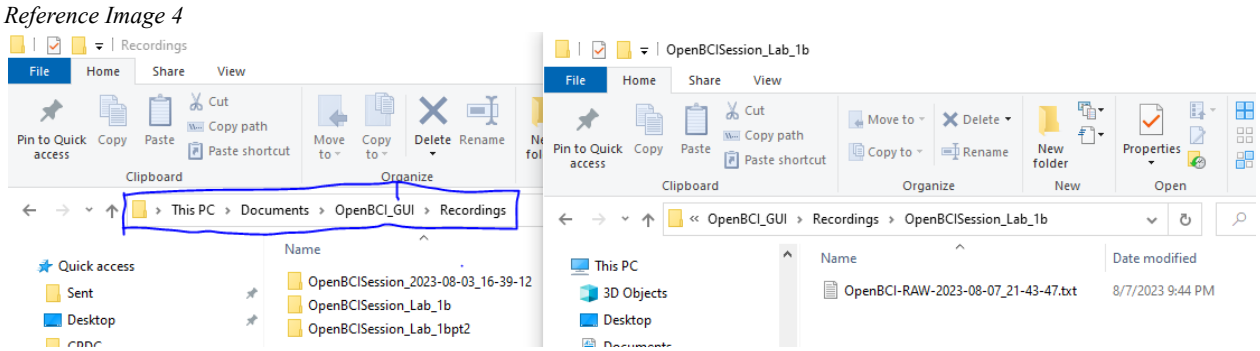
From now on, you can just click "Cyton (live)" and "Start Session", your board will remember its channel. Also, data files should automatically save for the full time you have data streaming, you don't need to specifically turn on file saving.

C) Finding the Data

The last thing we need to do is find the data file.

1) Close the OpenBCI GUI and then open it again, and start a data stream.
2) Record some kind of data for about a minute. A good thing to do is wiggle the Cyton board back and forth so the accelerometer will show that wiggling (you can see it in the GUI). Or, you can hook up EEG or ECG. The important thing is to just do something that you will be able to find when you plot the real data.
3) Stop the data stream, and close the software.
4) You should have a data file in your Documents folder (on the Mac it's /Users/USERNAME/Documents/OpenBCI_GUI/Recordings/). Find the session based on the time in the name of the folder. *(See Reference Image 4)*

Reference Image 4



⚠It is strongly recommended that the folder where the recordings are saved should not be on real time synchronization to the Cloud, as this can lead to discrepancies in the recorded data.⚠

In that session, there is a text file. The data file has a few "header" lines that look like this:

```
%OpenBCI Raw EEG Data
%Number of channels = 8
%Sample Rate = 250 Hz
%Board = OpenBCI_GUI$BoardCytonSerial
Sample Index, EXG Channel 0, EXG Channel 1, EXG Channel 2, EXG Channel 3, EXG Channel 4, EXG Channel 5, EXG Channel 6, EXG Channel 7, Accel Channel 0, Accel Channel 1, Accel Channel 2, Other, Other, Other, Other, Other, Other, Other, Analog Channel 0, Analog Channel 1, Analog Channel 2, Timestamp, Other, Timestamp (Formatted)
```

Then after that are rows of data, comma-separated. Each row is one sample from the system, with the corresponding data values.

⚠ The first number is the sample index, which will start at 0 and move up as long as you don't stop the data stream. If you stop the data stream and start again without closing the software, that index will restart at zero. This will be important to realize when you go to analyze your data. It's probably best to only start the data stream once, and then close the software after you stop it (to keep the files simple). ⚠

To read in the data in Matlab, use this bit of code:

```
dat = readtable('OpenBCI-RAW-2021-09-14_21-33-54.txt'); % insert your own filename
timedat = table2array(dat(:,25));
timedat = timedat-timedat(1);
timevals = seconds(timedat);
dat2 = table2array(dat(:,2));
plot(timevals,dat2);
```

Points to remember:

- Modify the filename
- Modify the channel number you're reading in (shown as channel 2 in this code) based on your data.
- Now, try to plot whatever data you recorded (e.g., the accelerometer). Generate a plot with the time values as the x-values, so that you can plot the full sample of data and see the signals you saw when recording.

⚠ You may notice, depending on the channel you plot, that it doesn't look quite like what was in the GUI. This is because the GUI showed filtered data, and you may be plotting the raw data. That's OK for now, we're going to apply the filters in software on the recorded data for future labs. ⚠

(For more information: Check the OpenBCI documentation for instructions on how to import the data files into Matlab and Python.
https://docs.openbci.com/Software/CompatibleThirdPartySoftware/Matlab/#using-the-openbci-gui)

⚠In every following lab ensure that you first make and check these basic Cyton connections and good signal collection (the accelerometer signal can be observed) before continuing with the experiment for the day.⚠

5) Result

In this lab you should have successfully connected to the OpenBCI board, allotted a specific channel, collected sample data, located the data and plotted the same on Matlab. If you've gotten this far, now you're ready to use the Cyton boards for experiments this semester!

6) Future Labs Prep

If you have spare time, you can try one of these setups to start preparing for future labs:

https://docs.openbci.com/GettingStarted/Biosensing-Setups/ECGSetup/
https://docs.openbci.com/GettingStarted/Biosensing-Setups/EEGSetup/
https://docs.openbci.com/GettingStarted/Biosensing-Setups/EMGSetup/